

Problems We Solve — Pattern Catalogue & Scope Guide

Four groups, nineteen sub-processes. Every pattern is the same shape, tagged by complexity, with an explicit scope boundary so we agree up front what we automate and what stays human.

Document: Internal pattern catalogue & scope guardrail

Companions: Discovery Reference · Scoping Worksheet · combined flow diagrams (PDF)

Version: v01

How to read this

The primitive. Almost every pattern below is one shape: Trigger → Capture → Classify / Decide → Draft or act, held for a human → Route · File · Log. Departments differ; the mechanics rarely do.

Complexity tiers. Base — capture, route, file/log with little branching. Base + triage — classify into a few classes plus a drafted reply held for review. Multi-branch — several decision points or checks before routing.

Scope boundary. Each sub-process states what VD automates and what stays human. It is a guardrail as much as a promise: it keeps us from being pulled past what we deliver, and it tells the client exactly where their judgement stays.

Cross-category by design. A request enters through one group but its steps are often served by sub-processes from several groups — e.g. an e-commerce enquiry may touch Communication & Intake, Order Exceptions and Returns at once. Place the request wherever the client sees it, then assemble the flow from whichever patterns fit.

What we don't do

VortexDeep automates recurring, supervised, moderate-volume work — where something arrives, a decision is made against clear rules, a person approves, and it is filed. We take on the automatable slice of a bigger problem (e.g. screening within hiring, or measurement within a quality programme) and say clearly where the human judgement stays. We do not do: autonomous decisions without a human; real-time or high-frequency systems; bespoke data-science / modelling; one-off data migrations; or consulting-led work such as running a Six Sigma or process-redesign project end to end.

A. Communication & Customers

1. Communication & Intake

Scope boundary — VD automates: capture, classification, drafted acknowledgements (held), routing, filing/logging · **Stays human:** approving replies, judgement on ambiguous enquiries

Example — Single inbox

Situation: Enquiries arrive by form, email and voicemail transcript; some are missed and replies are inconsistent.

Flow: Enquiry (any channel) → extract to a standard record → log to a central list → notify owner.

Signals: base — multi-channel intake, no classification, one human touchpoint

Diagram: *g1-intake-single-inbox*

Example — Language-and-type routing

Situation: A B2B supplier gets enquiries in DE/FR/EN and wants each routed with a courteous holding reply.

Flow: Enquiry → classify language + type → draft ack in sender's language (held) / route to support / file → route to salesperson.

Signals: base + triage — two classification dimensions, drafted reply held, routing to multiple owners

Diagram: *g1-intake-language-routing*

2. Customer Support

Scope boundary — VD automates: intent triage, KB-based draft answers (held), tagging, routing and escalation · **Stays human:** approving answers, resolving complex or non-standard issues, policy exceptions

Example — FAQ deflection

Situation: A shop's support inbox is swamped by repetitive order-status and returns questions.

Flow: Support email → classify intent → draft answer from KB (held) → agent approves & sends / escalate 'other'.

Signals: base + triage — 3 intent classes, KB lookup, drafted reply held, one escalation branch

Diagram: *g2-support-faq-deflection*

Example — Ticket enrichment & urgency

Situation: Critical tickets get buried; the team wants each tagged and the urgent ones surfaced.

Flow: Ticket → classify topic + severity → high: alert on-call + tag; normal: tag + queue → log.

Signals: multi-branch (light) — two dimensions, severity branch with alert, ticketing + alerting

Diagram: *g2-support-ticket-triage*

3. Scheduling & Booking

Scope boundary — VD automates: availability checks, drafted confirmations/alternatives (held), reminders, calendar writes · **Stays human:** approving sends, priority calls and special-case judgement

Example — Booking against the calendar

Situation: A practice gets booking requests out of hours; reception replies late or double-books.

Flow: Request → check availability → free: draft confirmation (held); not: draft alternatives (held) → send on approval + write calendar.

Signals: base + triage — calendar read/write, one availability branch, approval before send

Diagram: *g3-scheduling-booking*

Example — Reminders & no-shows

Situation: A clinic loses revenue to no-shows and wants reminders plus easy rescheduling.

Flow: Upcoming appointment → send reminder → change requested: flag reception + propose slot → update calendar on confirmation.

Signals: base — time-triggered, calendar integration, one reschedule branch

Diagram: *g3-scheduling-reminders*

11. Feedback & Reviews

Scope boundary — VD automates: sentiment/topic triage, drafted responses (held), routing, register logging · **Stays human:** approving replies, handling escalated complaints, product/service changes

Example — Sentiment triage & response

Situation: Feedback arrives scattered across email, review sites and surveys; praise is missed and complaints answered late.

Flow: Feedback → classify sentiment + topic → complaint: draft response + flag owner (held); praise: draft thanks (held); question: route to support → log to register.

Signals: base + triage — 3 classes, two drafted replies held, one route branch, register write

Diagram: *add3-feedback-reviews*

B. Sales & Growth

4. Lead & Sales Pipeline

Scope boundary — VD automates: enrichment, rule-based qualification, drafted intros/nudges (held), CRM logging · **Stays human:** qualification judgement on edge cases, the selling itself, deal decisions

Example — Lead qualification

Situation: Leads arrive from forms and events but sales chases everything equally.

Flow: Lead → enrich → qualify → hot: draft intro (held); nurture: list; not-a-fit: file → log to CRM.

Signals: base + triage — 3 classes, enrichment, drafted reply held, routing to 3 destinations

Diagram: *g4-pipeline-lead-qualification*

Example — Stale-deal nudges

Situation: Deals stall because reps forget to follow up.

Flow: Deal stalled → draft follow-up (held) → rep decision → send + log / move to lost + log.

Signals: base — time-triggered, CRM read/write, drafted message held, one decision branch

Diagram: *g4-pipeline-stale-deal-nudge*

5. Content & Marketing

Scope boundary — VD automates: draft generation, repurposing and idea triage (all held for review) · **Stays human:** editorial voice, approval, and what to publish (strategy)

Example — Repurposing content

Situation: A one-person marketing function has no time to turn a blog piece into social and newsletter copy.

Flow: Approved piece → generate channel drafts → hold all for review → approve/edit → file to content calendar.

Signals: base — single input to several drafted outputs (fan-out), one review gate

Diagram: g5-content-repurpose

Example — Content-idea triage

Situation: Recurring customer questions never become marketing content.

Flow: Question logged → classify (gap/covered/one-off) → gap: draft brief (held); else file → add approved briefs to backlog.

Signals: base + triage — 3 classes, drafted brief held, reads support/sales, writes backlog

Diagram: g5-content-idea-triage

6. E-commerce & Fulfilment

Scope boundary — VD automates: order screening, exception drafting (held), returns triage, routing/logging · **Stays human:** approving customer comms, refund/warranty decisions, judgement exceptions

Example — Order exceptions

Situation: Most orders are fine, but stock-outs, address issues and high-value orders need attention and slip through.

Flow: Order → screen stock/address/value → out-of-stock: draft notice (held); flag: human review; clear: fulfilment → log.

Signals: multi-branch — multiple checks, 3 outcome branches, one drafted notice held

Diagram: g6-ecom-order-exceptions

Example — Returns triage

Situation: The returns inbox mixes refunds, warranty claims and out-of-policy asks.

Flow: Return → classify → in-policy: draft approval (held); warranty: specialist; out-of-policy: draft decline (held) → log.

Signals: base + triage — 3 classes, two drafted replies held, specialist routing

Diagram: g6-ecom-returns-triage

12. Quotes & Proposals

Scope boundary — VD automates: requirement extraction, price/product matching, drafted quotes (held), routing and CRM logging · **Stays human:** pricing strategy, discount decisions, final approval

Example — Quote request to drafted quote

Situation: Sales hand-builds each quote from scattered price lists, so quotes go out late or with errors.

Flow: Request → extract requirements + match products/prices → complete: draft quote from template (held); missing info: draft clarification (held) → approve → send + log to CRM.

Signals: base + triage — extraction + price/product match, one completeness branch, drafted quote held, CRM write

Diagram: add1-quotes-proposals

13. Renewals & Retention

Scope boundary — VD automates: renewal tracking, risk flagging, drafted outreach (held), logging · **Stays human:** the retention conversation, discounts/commercial decisions, save strategy

Example — Renewal & win-back nudges

Situation: Renewals lapse or customers churn silently because nobody tracks upcoming renewals or early warning signs.

Flow: Renewal approaching / risk signal → assess account → healthy: draft reminder (held); at-risk: draft win-back + flag owner (held) → send on approval + update CRM.

Signals: base + triage — time + signal trigger, one health/risk branch, two drafted comms held, CRM read/write

Diagram: add6-renewals-retention

C. Documents & Finance

7. Document & Data Processing

Scope boundary — VD automates: extraction, classification, filing, drafted entries (held) • **Stays human:** verifying low-confidence extractions, handling non-standard documents

Example — Invoice / receipt extraction

Situation: Supplier invoices and receipts arrive as PDFs and photos and are retyped by hand.

Flow: Document → extract key fields → validate → confident: draft accounting entry (held); low: flag for manual keying → file + log.

Signals: base + triage — extraction from mixed formats, one confidence branch, entry held, accounting + store writes

Diagram: g7-docs-invoice-extraction

Example — Classification & filing

Situation: Contracts, invoices, delivery notes and post arrive in one place and are hard to find later.

Flow: Document → classify type + rename → confident: file to folder; uncertain: hold for human filing → log to index.

Signals: base + triage — 4 type classes, renaming, multi-destination filing, one uncertainty branch

Diagram: g7-docs-classify-file

8. Finance & Admin

Scope boundary — VD automates: gathering/matching, drafted payment lists and approvals (held), policy checks • **Stays human:** the money sign-off, exceptions and accounting judgement

Example — Payment-run preparation

Situation: Finance manually gathers the week's due invoices — slow and easy to mis-pay.

Flow: Payment-run day → gather due invoices → match to PO → matched: add to list; mismatch: flag → assemble list (held for approval) → mark scheduled + log.

Signals: base + triage — time-triggered, PO matching, one match branch, payment list held (money gate)

Diagram: g8-finance-payment-run

Example — Expense claims

Situation: Staff submit claims with mixed receipts; admin checks each against policy by hand.

Flow: Claim → check policy + receipts → within: draft approval (held); needs manager: route; missing receipt: draft request (held) → log.

Signals: base + triage — 3 classes, policy/receipt checks, two drafted comms held, manager routing

Diagram: *g8-finance-expense-claims*

14. Contracts & Agreements

Scope boundary — VD automates: term/date extraction, the contracts register and reminders · **Stays human:** legal interpretation, negotiation, the decision to renew or exit

Example — Key-date tracking

Situation: Signed contracts pile up and renewal or notice dates get missed, causing unwanted auto-renewals.

Flow: Contract filed → extract key terms + dates → dates found: write to register + schedule reminder; not: flag for human review → log.

Signals: base + triage — term/date extraction, one found/not-found branch, register write + reminder scheduling

Diagram: *add2-contracts-agreements*

17. Procurement & Suppliers

Scope boundary — VD automates: extraction, PO/delivery matching, drafted payment entries (held), flagging · **Stays human:** approving payment, resolving disputes, supplier negotiation and onboarding decisions

Example — Invoice-to-PO matching

Situation: Matching supplier invoices to POs and deliveries is manual, so overbilling and duplicate payments slip through.

Flow: Invoice → extract + match to PO / delivery → matched in tolerance: add to payment queue (held); mismatch / no PO: flag to buyer → log.

Signals: base + triage — extraction + 3-way match, one match/mismatch branch, payment entry held, buyer escalation

Diagram: *add7-procurement-suppliers*

18. Compliance & Filing

Scope boundary — VD automates: gathering, assembly, consistency checks, deadline reminders, audit-trail archiving · **Stays human:** the filing sign-off, tax judgement and formal responsibility (accountant / fiduciary)

Example — VAT return preparation

Situation: An SME scrambles each period to assemble VAT figures and nearly misses filing deadlines.

Flow: Filing period due → gather transactions + assemble draft return → checks pass: hold draft for review; anomaly: flag → on approval: submit + archive audit trail.

Signals: base + triage — time-triggered, assembly, one checks branch, return held for approval (compliance gate), audit-trail write

Diagram: *add8-compliance-filing*

D. Operations & Insight

9. Internal Operations

Scope boundary — VD automates: checklist creation, task assignment, notifications, tracking, reminders, request routing · **Stays human:** doing the tasks themselves, decisions on exceptions and approvals

Example — Onboarding coordination

Situation: A new joiner needs IT, equipment, access and paperwork kicked off across several people; steps get missed.

Flow: New hire → create checklist + assign owners → notify + track → overdue: nudge; else confirm → mark done + log.

Signals: base — multi-owner assignment, notifications, tracking with an overdue-reminder branch

Diagram: g9-ops-onboarding

Example — Internal request handling

Situation: Ad-hoc access, equipment and facilities requests get lost in a shared inbox.

Flow: Request → classify → route to owner + draft ack (held) → owner actions → log / close.

Signals: base + triage — 3+ request classes, routing to owners, drafted acknowledgement held

Diagram: g9-ops-internal-requests

10. Reporting & Analytics

Scope boundary — VD automates: data pulls, standard assembly, sanity checks, drafted reports (held), threshold alerts · **Stays human:** interpretation, decisions from the numbers, root-cause analysis

Example — Recurring report assembly

Situation: Someone spends hours each cycle pulling the same numbers into a management report.

Flow: Reporting day → pull figures → assemble standard report → checks pass: hold draft; anomaly: flag → publish on approval + file.

Signals: base + triage — scheduled, multi-source pulls, one checks branch, draft held, publish + file

Diagram: g10-reporting-recurring

Example — Threshold / exception alerting

Situation: Problems are found too late because nobody watches the numbers continuously.

Flow: New data → compare to thresholds → within: log; breach: compile alert → notify owner + log.

Signals: base + triage (monitoring) — periodic monitoring, threshold comparison, internal alert, owner notify

Diagram: g10-reporting-alerting

15. People & Hiring

Scope boundary — VD automates: acknowledgement, criteria screening, interview scheduling, status tracking, keep-warm nudges · **Stays human:** shortlisting judgement, interviews and the hiring decision

Example — Application screening & scheduling

Situation: A posted role floods the inbox; manual screening and scheduling is slow and good candidates go cold.

Flow: Application → screen against must-have criteria → meets: shortlist + propose interview slots (held); below: draft polite decline (held) → log + track status.

Signals: base + triage — rule-based screen, one qualify branch, two drafted comms held, scheduling + tracking

Diagram: *add4-people-hiring*

16. Quality & Process Monitoring

Scope boundary — VD automates: the Measure & Control layer — data capture, metric computation, control-limit monitoring, exception alerts, trend packs · **Stays human:** Define / Analyse / Improve — problem definition, root-cause analysis, process redesign and improvement decisions

Example — Measure & control layer

Situation: A quality / continuous-improvement effort needs steady measurement, but data is gathered by hand and problems surface late.

Flow: New process data (periodic) → compute metrics vs control limits → within: log to trend pack; out: compile exception with context → notify process owner + log.

Signals: base + triage (monitoring) — periodic collection, metric + limit comparison, exception compile, owner notify

Diagram: *add5-quality-monitoring*

19. Knowledge & SOPs

Scope boundary — VD automates: search, drafted answers, SOP-stub drafting (held), publishing · **Stays human:** authoring authoritative content, approving accuracy, deciding what's canonical

Example — Internal FAQ / SOP upkeep

Situation: Staff repeatedly ask the same 'how do we do X'; answers live in people's heads and scattered docs.

Flow: Question → search existing SOPs / knowledge → found: draft reply from source (held); gap: draft SOP stub + flag owner (held) → publish / send on approval + log.

Signals: base + triage — KB search, one found/gap branch, drafted reply or SOP stub held, KB write

Diagram: *add9-knowledge-sops*